

100 Days of Azure DevOps

Day 3 Handout — Control plane, tags & locks

Learning in public · Personal lab only · Educational content · Not a sales pitch

What you will understand today

- **ARM** is the control plane behind Portal, CLI, and PowerShell
- **Hierarchy** — management group → subscription → resource group → resource
- **Tags** — sticky notes for cost and ownership
- **Locks** — duct tape so prod is harder to delete by accident

High-level architecture

Architecture A — request path

Layer	What it does
You (Portal / CLI / PowerShell)	Express the change you want
Azure Resource Manager (ARM)	Auth, validate, orchestrate the request
Resource providers	Create/update the real resource (Storage, Web, ...)
Resource	The thing that exists and (often) costs money

Architecture B — scope hierarchy

Level	Metaphor	Why it matters
Management group	Company / folder of folders	Policy at scale
Subscription	Billing boundary	Where invoices land
Resource group	Project box	Lifecycle + delete scope
Resource	The actual thing	VM, storage, web app, ...

Architecture C — tags vs locks

Tool	Job
Tags	Label owner / project / env for cost and cleanup
CanNotDelete lock	Block delete until lock is removed
ReadOnly lock	Block most changes (stricter)

One-liner to remember

Portal/CLI = how you talk · ARM = who enforces rules · Tags = sticky notes · Locks = duct tape

Step-by-step lab (20–30 min)

Personal subscription only. Delete the lab RG when finished.

- 1 Create **rg-day03-lab** with tags Project=100Days, Owner=personal, Env=lab
- 2 Add a **CanNotDelete** lock on the resource group
- 3 Try **az group delete** — expect failure while locked

4 Delete the lock, then delete the resource group

Commands

```
az group create -n rg-day03-lab -l centralindia --tags Project=100Days Owner=personal Env=lab
az lock create --name cannot-delete --lock-type CanNotDelete --resource-group rg-day03-lab
az lock delete --name cannot-delete --resource-group rg-day03-lab
az group delete -n rg-day03-lab --yes --no-wait
```

Done checklist

- I can explain ARM in one sentence
- I applied tags and a CanNotDelete lock
- I cleaned up rg-day03-lab

Tomorrow — Day 4

DevOps principles & culture — CALMS and DORA without the buzzword fog.

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